

The solutions for this sheet must be submitted on Moodle by 16 April, 15:59. Afterwards you are assigned solutions of another student, which you have to grade (and upload to Moodle) until 11:59 of the following TUESDAY.

Exercise P3.1 – Distributions

- (a) While sending a message $m \in \{0, 1\}^n$, some errors may occur. Each of the n bits is flipped with probability p . All flips happen independently of each other. What is the probability that exactly k bits are flipped?
- (b) Alice and Bob take turns tossing coins. The person to get “tails” first wins. Since Alice begins, Bob uses a biased coin. This biased coin shows “tails” with probability $1/2 < p < 1$. Alice has a fair coin. What is Alice’s probability of winning? Can p be chosen such that the game is fair (i.e., both players have the same probability of winning)?
- (c) You and your friend want to play frisbee. Unfortunately, neither of you owns a frisbee. Instead you take n plates from your WG’s kitchen. The two of you start with the first plate and throw it back and forth between each other. For every throw there is a $\frac{1}{2}$ chance that the plate is caught and a $\frac{1}{2}$ chance that the plate drops and hence breaks. Once a plate breaks, you take the next one until all plates are broken. All throws are independent of each other. Let X be the random variable counting the number of throws that are caught. Similarly, let Y be the random variable counting the number of throws that are not caught. Determine the expected value and the variance of X , Y , and $X + Y$.

Hint: It might be easier to start with Y and $X + Y$. You may use without proof that for independent variables $A_1, \dots, A_k$ we have $\text{Var}[\sum_{i=1}^k A_i] = \sum_{i=1}^k \text{Var}[A_i]$.